Brady M. Chisholm

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Research Interests

I am interested in how listeners extract meaningful signals from competing sound, and how that ability changes with hearing loss, age, or auditory prostheses. I have approached this question at several levels: spatial and comodulation masking release in an animal model, speech-in-speech perception and listening effort in human listeners with and without mild cognitive impairment, and pitch coding in cochlear-implant and auditory-nerve-implant recipients. I pair behavioral psychophysics with physiological measures — EEG and pupillometry — and analyze the resulting data with mixed-effects and generalized additive models in R and MATLAB.

Education

Bachelor of Science, Psychology *Sept. 2023–Aug. 2025*

University of Minnesota, Minneapolis, Minnesota

- Awarded scholarship and completed Dean's First-Year Research and Creative Scholars, training in Professor Andrew Oxenham's lab in data collection for EEG and auditory-based experiments, and audiogram methods
- Quantitative & computational: Regression & Correlated Data; R & Reproducible Research; Introduction to Programming Concepts (Python); Calculus I; Introduction to Physics
- Domain: Psychology of Hearing; Psychology of Attention; Research Methods

Gustavus Adolphus College, St. Peter, Minnesota *Sept. 2022–May 2023*

Coursework toward B.A.; transferred to UMN

- Completed 2 semesters of Research Apprenticeship, mentored by Professor Lauren Hecht on designing an experiment investigating entomophobic responses to the Madagascar hissing cockroach; in collaboration with Gustavus Biology Department
- Varsity Men's Swim & Dive Competitor, MIAC Conference

Research Experience

Research Professional *August 2026–Present*

Sensory Optimization & Neural Implant Coding Lab, UMN Dept. of Biomedical Engineering (PI: Dr. Hubert Lim)

- Validate a six-module psychophysical test suite (MATLAB) for an in-human auditory nerve implant (ANI), verifying stimulus output and response hardware through the MED-EL research stimulation interface and Picoscope measurement prior to clinical release.
- Deliver verified protocols to the collaborating clinical team in Hannover, Germany, for administration with the first implanted ANI recipient.
- Lead recruitment and data collection for a pitch-scaling study in normal-hearing and cochlear-implant listeners, establishing comparison data for the same task in ANI recipients.
- Build independent analysis pipelines for pitch-scaling data in MATLAB.

Research Professional (25% appointment) *August 2026–Present*

Center for Applied Translational and Sensory Science, UMN College of Liberal Arts (Director: Dr. Andrew Oxenham)

- Administer free hearing and vision screenings aboard the See, Hear, MN mobile sensory unit, a sound-attenuating screening suite serving underserved metro and rural Minnesota communities; over 1,000 screenings completed to date.
- Operate and deploy the unit to county fairs, health festivals, food pantries, and the Minnesota State Fair, including site setup enabling data collection by partner CATSS labs.
- Counsel community members on sensory health and referral pathways for follow-up care.

Psychology Consultant *May 2026–July 2026*

Dr. Piper Ranallo, UMN Institute for Health Informatics

- Conducted a structured review of over 300 behavioral health assessment instruments, tracing each to its original development publication and subsequent validation studies.
- Built and delivered a structured database cataloging instrument provenance, psychometric validation, and source citations.

Research Professional *November 2025–July 2026*
Research Assistant *May 2024–August 2025*
Auditory Perception & Cognition Lab, UMN Psychology Department (PI: Dr. Andrew Oxenham)

- Assisted with NIH funded research investigating human speech and speech-in-speech perception via EEG and behavioral experiments.
- Recruited MCI (mild cognitive impairment) population through M Health Fairview for listening based EEG experiments investigating listening differences between MCI and normal hearing listeners.
- Processed, analyzed, and visualized multimodal datasets in R and MATLAB; applied GAMMs and mixed-effects modeling.

Research Assistant *May 2025–October 2025*
Noel Lab, UMN Neuroscience Department (PI: Dr. Jean-Paul Noel)

- Designed, recruited and piloted human participants (N = 11) independently in an experiment investigating visual encoding and reward prediction, managed data collection and preprocessing.
- Built analysis pipelines in R and MATLAB using signal filtering; synchronizing data.
- Organized and implemented lab-wide human participant database for HIPAA and IRB compliance.

Research Assistant *May 2024–August 2024*
Animal Communication Lab, UMN Ecology, Evolution, and Behavior (PI: Dr. Mark Bee)

- Supported NSF-funded research on spatial release from masking (SRM) and comodulation masking release (CMR) in *Hyla chrysoscelis*, an animal model of signal detection in noisy, multi-talker environments (acknowledged contributor)
- Conducted two-alternative phonotaxis tests across six noise treatments factorially varying noise location (colocated vs. spatially separated) and envelope modulation (unmodulated, comodulated, uncorrelated)
- Scored response, response latency, and stimulus selection; assisted with acoustic calibration of the testing arena
- Performed animal handling, surgical preparation, and post-surgical care under IACUC protocols; collected field specimens during the spring breeding season

Publications, Posters, and Presentations

Publications

1. **Chisholm, B. M.**, Johns, M. A., & Mesik, J. *Fatigue Does Not Interact with Experimental Manipulation in a Speech-in-Noise Listening Effort Task.* (Manuscript in preparation).

Posters and Presentations

1. **Chisholm, B. M.**, Johns, M. A., & Mesik, J. (2026, April). *Assessing the Effect of Pupillometric Fatigue on Task-Evoked Responses in a Listening Effort Paradigm.* [Poster Presentation]. Center for Applied & Translational Sensory Science Spring Research Day, University of Minnesota, Minneapolis, MN.

2. **Chisholm, B. M.**, & Mesik, J. (2025, May). *Assessing the Effect of Pupillometric Fatigue on Task-Evoked Responses in a Listening Effort Paradigm.* [Poster Presentation]. 188th Meeting of the Acoustical Society of America, New Orleans, LA.

3. **Chisholm, B. M.**, & Hecht, L. (2023, May). *Impact of a Cockroach in the Rubber Hand Illusion.* [Oral Presentation]. Spring Research Day, Gustavus Adolphus College, St. Peter, MN.

Honors, Awards, and Scholarships

College of Liberal Arts Dean's List, *University of Minnesota* *2024-2025*

1st Place, R Data Science Competition, *Psi Chi International Honor Society* *Oct. 2024*

Undergraduate Research Opportunities Program, *University of Minnesota* *Summer 2024*

Dean's First-Year Research & Creative Scholars, *University of Minnesota* *Spring 2024*

Director's Residential Service Award, *Supportive Living Solutions* *February 2024*

President's Faculty-Student Collaboration Grant, *Gustavus Adolphus College* *Summer 2023*

Dean's List, *Gustavus Adolphus College* *Spring 2023*

Swanson-Holcomb Undergraduate Research Award, *Gustavus Adolphus College* *Spring 2023*

Eagle Scout, *Northern Star Council, Boy Scouts of America* *Nov. 2021*

Skills & Certifications

Lab & Technical Skills

- EEG Data Collection and Post-Processing (BioSemi; EEGLab)
- Pupillometry (Tobii Pro Spectrum)
- Oscilloscope/Picoscope Testing
- Audiometric Threshold Testing
- Surgical Support & Animal Handling
- Laboratory Safety & Data Protocols; GCP

Programming & Tools

- **Proficient:** MATLAB, R, Markdown, Git/GitHub
- **Modeling:** GAMMs, linear mixed-effects, regression, non-parametric statistics
- **R packages:** *ggplot2*, *mgcv*, *itsadug*, *lme4*, *tidyverse*
- **Familiar:** Python, HTML, LaTeX, Linux OS

Certifications

- Fairview Non-employed Research Staff (human subjects research through M Health Fairview)
- Collaborative Institutional Training Initiative (CITI) Training (data practices, research privacy & ethics)
- HIPAA, Information Security Training
- MoCA (Montreal Cognitive Assessment) Test Administration Certification for Mild Cognitive Impairment

Service & Leadership

Reviewing

Ad hoc reviewer, *Minnesota Undergraduate Research & Academic Journal* (2024-2025)

Other Professional Service

Appointed Member, AI Research Committee *Sept 2026–Present*
University of Minnesota, Department of Psychology

- Departmental task force dedicated to ensuring the department effectively and appropriately leverages AI within the research lifecycle

Committee Lead, AI in Research Task Force *Feb 2026–May 2026*
University of Minnesota, Department of Psychology

- Led a departmental task force charged with establishing ethical guidelines for Generative AI use in the Psychology Department's research workflow.
- Authored a group report on data privacy, mentorship, and AI-assisted research.

Academic Integrity Matters (AIM) Program Volunteer *Jan 2026–Present*
Office of Community Standards, UMN Twin Cities

- Assist with facilitating discussions with students who violated the University Student Code of Conduct; program aimed towards restorative justice.
- Represent University Community Members, tasked with promoting academic integrity by sharing perspectives on the impact of scholastic dishonesty beyond the classroom.

Science Outreach & Communication

Community Presentation, Orchestra Hall: Music and the Mind *Feb. 15, 2026*
UMN Auditory Perception & Cognition Lab, Minneapolis, MN

- Provided information on auditory science and research for Cochlear Implant (CI) users and broader community during a chamber concert event.

Guest Speaker, UMN CLA 1052 *Fall 2025*
University of Minnesota, Minneapolis, MN

- Served as panelist on undergraduate research pathways and opportunities for engagement; within and beyond the psychology department.

Community Presentation, Spotlight Science: The Science of Sound *Jan. 18, 2025*
Bell Museum, St. Paul, MN

- Assisted with interactive demonstrations explaining the auditory system to the public.

Additional Experience

Mental Health Advocate II *May 2023–August 2024*
Supportive Living Solutions Whittier Campus, Minneapolis, Minnesota

- Coordinated care for 40+ residents with diverse behavioral and mental health needs with a multidisciplinary clinical team.
- Audited and managed medication systems to ensure full compliance with state and federal regulations.
- Implemented improvements to documentation accuracy and continuity of care via tracking system.

Lead Copy Editor *September 2022–June 2023*
The Gustavian Weekly, St. Peter, Minnesota

- Managed a team of three copy editors in reviewing, editing, and publishing a weekly print newspaper.